

POST-QUANTUM CRYPTOGRAPHY (PQC)

Understanding PQC & The 'Harvest Now, Decrypt Later' Hacker Threat

Plain-English & Detailed Technical Guide | 26-August-2026

Why are companies upgrading to Post-Quantum Cryptography today when Quantum Computers are still being built? Because of a major cyber threat called 'Harvest Now, Decrypt Later' (HNDL).

STAGE 1: WHAT HACKERS ARE DOING TODAY (HARVESTING)

Hackers, hostile states, and cybercriminals are tapping internet cables, capturing database backups, and recording encrypted network traffic TODAY. They save all these encrypted files onto giant storage drives. They cannot open them today because regular computers would take 10,000 years to crack the lock.

STAGE 2: WHAT HACKERS WILL DO TOMORROW (DECRYPTING)

In a few years, when Quantum Computers are built, hackers will plug those saved files into the Quantum Computer. Using Shor's Algorithm, the Quantum Computer will pop open standard RSA/ECC locks in SECONDS, exposing all passwords, bank records, medical histories, and trade secrets.

2. THE SIMPLE DIGITAL LOCK INSPECTOR ANALOGY

Think of our software as an automated Digital Lock Inspector. When you upload server files, it inspects all your digital locks and color-codes your security risk into 3 simple tiers:

RED LOCKS (VULNERABLE) - Quantum-Breakable Standard Locks

RSA-2048/4096, ECC P-256/384, ECDSA, Diffie-Hellman.
Must be replaced to prevent future decryption.

ORANGE LOCKS (WEAK) - Classically Insecure & Deprecated Locks

3DES, DES, RC4, MD5, SHA-1.
Weak locks that are already risky today.

GREEN LOCKS (SAFE) - NIST Quantum-Proof Locks

AES-256, SHA-384, ML-KEM-768, ML-DSA, SLH-DSA.
Strong locks that Quantum Computers can NEVER break.

3. DETAILED TECHNICAL EXPLANATION (How Live Data Parsing Works)

Are we guessing or using dummy data? NO. Our PQC Engine reads 100% REAL, live data directly out of the actual server configuration files, X.509 certificates, and network configs uploaded by the user.

Step 1: Reading Live File Parameters

When you upload `sql.config`, `nginx.conf`, or a `.crt` certificate, our parser reads the actual text settings configured on the client's real server (e.g. `ssl_ciphers ECDHE-RSA-AES128-GCM-SHA256`, `tls_version = TLSv1.3`, or X.509 OID `1.2.840.113549.1.1.1`).

Step 2: Mathematical Quantum Evaluation

The parser compares each extracted algorithm against NIST FIPS 203 (ML-KEM), FIPS 204 (ML-DSA), and FIPS 205 (SLH-DSA) rules:

- * Shor's Algorithm Rule: Any algorithm solving prime factoring (RSA) or discrete logs (ECC/DH) is evaluated as VULNERABLE.
- * Grover's Algorithm Rule: Symmetric ciphers (AES-256) retain 128-bit quantum resistance and are evaluated as SAFE.

Step 3: Cryptographic Bill of Materials (CBOM) Generation

Every extracted finding is structured into a CBOM record containing the exact asset name, algorithm, protocol version, exposure tier, and severity score.

Step 4: Automated Remediation Mapping

The engine maps each vulnerability to copy-paste remediation commands (e.g. `tls_version = TLSv1.3`, enable OpenSSL 3.4+ PQC provider) and outputs executive PDF & Word reports.

4. DEMO SCRIPT & CLIENT Q&A CHEAT SHEET

30-Second Opening Pitch for Tomorrow's Demo:

"Good morning / afternoon. Today we are demonstrating the AI Cyber Audit Box Post-Quantum Cryptography (PQC) Engine. Our software automatically inspects a company's server configuration files and certificates, identifies all vulnerable digital locks that future quantum computers can break using Shor's algorithm, and delivers a 1-click migration roadmap to NIST Quantum-Safe Standards."

Quick Answers to Client Questions during the Demo:

Q: Why do we need PQC if Quantum Computers aren't here yet?

A: Because hackers are harvesting encrypted data TODAY and saving it on hard drives to decrypt later when Quantum Computers arrive ('Harvest Now, Decrypt Later').

Q: Do we need to share private keys with your scanner?

A: NO. The scanner parses public certificates (.pem/.crt) and server configuration parameters. Private keys are never required or exposed.

Q: How fast does the scan run?

A: Sub-second execution for configuration files and certificates. Multi-server enterprise batch audits complete in under 5 seconds.

Q: Does your tool align with official NIST standards?

A: YES. Full coverage of finalized NIST standards: FIPS 203 (ML-KEM), FIPS 204 (ML-DSA), and FIPS 205 (SLH-DSA).

AI CYBER AUDIT BOX -- PQC ENGINE DEMO GUIDE READY